

# HERVE JUNIOR KAMMEGNE

Red Deer, Alberta T4R 3K9, Canada | (403)-307-5509 | [kammegneherve@gmail.com](mailto:kammegneherve@gmail.com) | [LinkedIn](#) | [GitHub](#)

## PROFILE

Motivated Junior Data Analyst and Artificial Intelligence student at Red Deer Polytechnic with hands-on experience in data analysis, machine learning, and predictive modeling. Strong foundation in Python, SQL, statistical analysis and neural network with practical academic projects in photovoltaic damage detection, fraud detection, and data pipeline development. Passionate about applying AI and analytics to real-world problems in energy innovation and sustainable technologies.

## EDUCATION

Red Deer Polytechnic

Artificial Intelligence & Data Analytics Program

Aug 2024 to May 2026

Relevant Coursework:

- Machine Learning & Predictive Modeling
- Data Analytics & Visualization
- Database Design & SQL Statistical Analysis
- Python Programming for Data Science
- Data Cleaning & Feature Engineering
- Neural network & Deep Learning
- Computer Vision Linear Algebra

Red Deer Polytechnic

Academic Upgrading (ENG095)

May 2024

## TECHNICAL SKILLS

|                                                    |                      |                     |                       |
|----------------------------------------------------|----------------------|---------------------|-----------------------|
| • Python (Pandas, NumPy, Scikit-learn, Matplotlib) | • SQL                | • Power BI          | • Excel               |
| • ETL pipelines                                    | • Data Cleaning      | • Schema Design     | • Data Normalization  |
| • Regression                                       | • Classification     | • Model Evaluation  | • Feature Engineering |
| • Statistical analysis                             | • Dashboard creation | • Data storytelling | • MLflow              |
| • DVC                                              | • AWS                | • Dagshub           | • Tesseract OCR       |
| • OpenCV                                           | • Yolo               | • Git               | • VS Code             |
| • Jupyter Notebook                                 | • Anaconda Navigator |                     |                       |

## ACADEMIC & TECHNICAL PROJECTS

- **Solar Panel photovoltaic damage detection (Convolution task)**  
  
Analyzed solar photovoltaic datasets from Kaggle. Built a convolution model mixed to classification model to detect the different damage and classify them into different group crack panels or not. Applied statistical analysis to identify optimization opportunities.
- **Credit Card Fraud Detection (Machine Learning Project)**  
  
Preprocessed large-scale transactional dataset. Addressed class imbalance and engineered predictive features. Developed classification models and evaluated using precision, recall, and ROC-AUC. Improved detection accuracy through hyperparameter tuning.
- **Survey Design, Analysis, and Classification Project**

Designed Project Charter. Designed a Survey. Create a Code to process the survey results, Explore Data and Clean Responses. Establish Classification Model.

- **Graph-Based Optimization Project**

Modeled real-world routing problems using weighted graphs. Implemented Dijkstra’s algorithm and A\* search for route optimization. Conducted performance comparisons and fuel cost estimation simulations.

- **License Plate Detection & Recognition System**

Developed a complete license plate detection and recognition pipeline using classical computer vision techniques (no deep learning models). Applied image preprocessing techniques including grayscale conversion, Gaussian blur, and Canny edge detection. Implemented contour detection and geometric filtering (aspect ratio constraints) to localize plate regions. Extracted and cropped plate images using bounding box coordinates. Applied Otsu thresholding and integrated Tesseract OCR (tesseract) for character recognition. Tested the full pipeline on real-world vehicle images.

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## STUDENT EXPERIENCE

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|----------------------------------------------------|----------------------------|
| <b>Crew Member / Cashier, Jollibee</b><br>Red Deer | <b>Mar 2025 to Present</b> |
|----------------------------------------------------|----------------------------|

- Managed transactional data and cash handling with high accuracy.
- Worked in a fast-paced environment requiring analytical thinking and efficiency.
- Collaborated in team-based operations to optimize service performance.

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|-------------------------------------------------------------------------|----------------------------|
| <b>Kitchen Helper / Dishwasher, Westerner Park Centrium</b><br>Red Deer | <b>Jan 2024 to Present</b> |
|-------------------------------------------------------------------------|----------------------------|

- Maintained high standards of organization and workflow efficiency.
- Demonstrated strong time management under pressure.

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| <b>Volunteer, Red Deer Food Bank</b> | <b>May 2024 to Present</b> |
|--------------------------------------|----------------------------|

- Ensured compliance with food safety procedures.
- Assisted with inventory organization and logistics coordination.

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| <b>Tutor, Francophonie Albertaine Plurielle (FRAP)</b> | <b>Apr 2024 to Aug 2024</b> |
|--------------------------------------------------------|-----------------------------|

- Supported students in structured learning environments.
- Developed communication and problem-solving skills.

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## KEY COMPETENCIES AND PERSONAL SKILLS

- Data Analysis & Interpretation**
- Machine Learning Fundamentals**
- Problem Solving & Critical Thinking**
- Research & Analytical Mindset**
- Team Collaboration**
- Strong Work Ethic**
- Bilingual: English & French**

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## LINKS

[GitHub](#), [LinkedIn](#)

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## LANGUAGES

English, French